

Product Support Manual

Veritas Forge Industries Ltd

CHANAKYA Demonstration Dataset - synthetic data for a fictional company. Not real.

1. Using this Manual

Each product section lists its specification and common fault codes. Escalate to the field service desk if a fault persists after the recommended action.

2. Product FC-110

Product FC-110 belongs to the Forge Controller series. Warranty is 12 months from commissioning.

2.1 Specification FC-110

Parameter	Value
Model	FC-110
Warranty (months)	12
Rated Voltage (V)	415
Service Interval (months)	6

2.2 Troubleshooting FC-110

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
FC-110-E01	intermittent shutdown	Faulty encoder	Replace or reseal encoder and retest	P2
FC-110-E02	error code on display	Faulty servo drive	Replace or reseal servo drive and retest	P3
FC-110-E03	abnormal noise	Faulty power supply	Replace or reseal power supply and retest	P3
FC-110-E04	loss of calibration	Faulty firmware	Replace or reseal firmware and retest	P4
FC-110-E05	overheating	Faulty coolant pump	Replace or reseal coolant pump and retest	P2
FC-110-E06	no response to controls	Faulty display	Replace or reseal display and retest	P3

3. Product FC-120

Product FC-120 belongs to the Forge Controller series. Warranty is 18 months from commissioning.

3.1 Specification FC-120

Parameter	Value
Model	FC-120
Warranty (months)	18
Rated Voltage (V)	230
Service Interval (months)	9

3.2 Troubleshooting FC-120

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
FC-120-E01	error code on display	Faulty firmware	Replace or reseal firmware and retest	P2
FC-120-E02	abnormal noise	Faulty coolant pump	Replace or reseal coolant pump and retest	P3
FC-120-E03	loss of calibration	Faulty display	Replace or reseal display and retest	P3
FC-120-E04	overheating	Faulty sensor array	Replace or reseal sensor array and retest	P4
FC-120-E05	no response to controls	Faulty gearbox	Replace or reseal gearbox and retest	P2
FC-120-E06	intermittent shutdown	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3

4. Product FC-210

Product FC-210 belongs to the Forge Controller series. Warranty is 24 months from commissioning.

4.1 Specification FC-210

Parameter	Value
Model	FC-210
Warranty (months)	24
Rated Voltage (V)	415
Service Interval (months)	12

4.2 Troubleshooting FC-210

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
FC-210-E01	abnormal noise	Faulty sensor array	Replace or reseal sensor array and retest	P2
FC-210-E02	loss of calibration	Faulty gearbox	Replace or reseal gearbox and retest	P3
FC-210-E03	overheating	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3
FC-210-E04	no response to controls	Faulty cable harness	Replace or reseal cable harness and retest	P4
FC-210-E05	intermittent shutdown	Faulty encoder	Replace or reseal encoder and retest	P2
FC-210-E06	error code on display	Faulty servo drive	Replace or reseal servo drive and retest	P3

5. Product FC-220

Product FC-220 belongs to the Forge Controller series. Warranty is 12 months from commissioning.

5.1 Specification FC-220

Parameter	Value
Model	FC-220
Warranty (months)	12
Rated Voltage (V)	230
Service Interval (months)	15

5.2 Troubleshooting FC-220

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
FC-220-E01	loss of calibration	Faulty cable harness	Replace or reseal cable harness and retest	P2
FC-220-E02	overheating	Faulty encoder	Replace or reseal encoder and retest	P3
FC-220-E03	no response to controls	Faulty servo drive	Replace or reseal servo drive and retest	P3
FC-220-E04	intermittent shutdown	Faulty power supply	Replace or reseal power supply and retest	P4
FC-220-E05	error code on display	Faulty firmware	Replace or reseal firmware and retest	P2
FC-220-E06	abnormal noise	Faulty coolant pump	Replace or reseal coolant pump and retest	P3

6. Product FC-310

Product FC-310 belongs to the Forge Controller series. Warranty is 18 months from commissioning.

6.1 Specification FC-310

Parameter	Value
Model	FC-310
Warranty (months)	18
Rated Voltage (V)	415
Service Interval (months)	6

6.2 Troubleshooting FC-310

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
FC-310-E01	overheating	Faulty power supply	Replace or reseal power supply and retest	P2
FC-310-E02	no response to controls	Faulty firmware	Replace or reseal firmware and retest	P3
FC-310-E03	intermittent shutdown	Faulty coolant pump	Replace or reseal coolant pump and retest	P3
FC-310-E04	error code on display	Faulty display	Replace or reseal display and retest	P4
FC-310-E05	abnormal noise	Faulty sensor array	Replace or reseal sensor array and retest	P2
FC-310-E06	loss of calibration	Faulty gearbox	Replace or reseal gearbox and retest	P3

7. Product FC-410

Product FC-410 belongs to the Forge Controller series. Warranty is 24 months from commissioning.

7.1 Specification FC-410

Parameter	Value
Model	FC-410
Warranty (months)	24
Rated Voltage (V)	230
Service Interval (months)	9

7.2 Troubleshooting FC-410

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
FC-410-E01	no response to controls	Faulty display	Replace or reseal display and retest	P2
FC-410-E02	intermittent shutdown	Faulty sensor array	Replace or reseal sensor array and retest	P3
FC-410-E03	error code on display	Faulty gearbox	Replace or reseal gearbox and retest	P3
FC-410-E04	abnormal noise	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P4
FC-410-E05	loss of calibration	Faulty cable harness	Replace or reseal cable harness and retest	P2
FC-410-E06	overheating	Faulty encoder	Replace or reseal encoder and retest	P3

8. Product HP-110

Product HP-110 belongs to the Hydraulic Press series. Warranty is 12 months from commissioning.

8.1 Specification HP-110

Parameter	Value
Model	HP-110
Warranty (months)	12
Rated Voltage (V)	415
Service Interval (months)	12

8.2 Troubleshooting HP-110

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
HP-110-E01	intermittent shutdown	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P2
HP-110-E02	error code on display	Faulty cable harness	Replace or reseal cable harness and retest	P3
HP-110-E03	abnormal noise	Faulty encoder	Replace or reseal encoder and retest	P3
HP-110-E04	loss of calibration	Faulty servo drive	Replace or reseal servo drive and retest	P4
HP-110-E05	overheating	Faulty power supply	Replace or reseal power supply and retest	P2
HP-110-E06	no response to controls	Faulty firmware	Replace or reseal firmware and retest	P3

9. Product HP-120

Product HP-120 belongs to the Hydraulic Press series. Warranty is 18 months from commissioning.

9.1 Specification HP-120

Parameter	Value
Model	HP-120
Warranty (months)	18
Rated Voltage (V)	230
Service Interval (months)	15

9.2 Troubleshooting HP-120

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
HP-120-E01	error code on display	Faulty servo drive	Replace or reseal servo drive and retest	P2
HP-120-E02	abnormal noise	Faulty power supply	Replace or reseal power supply and retest	P3
HP-120-E03	loss of calibration	Faulty firmware	Replace or reseal firmware and retest	P3
HP-120-E04	overheating	Faulty coolant pump	Replace or reseal coolant pump and retest	P4
HP-120-E05	no response to controls	Faulty display	Replace or reseal display and retest	P2
HP-120-E06	intermittent shutdown	Faulty sensor array	Replace or reseal sensor array and retest	P3

10. Product HP-210

Product HP-210 belongs to the Hydraulic Press series. Warranty is 24 months from commissioning.

10.1 Specification HP-210

Parameter	Value
Model	HP-210
Warranty (months)	24
Rated Voltage (V)	415
Service Interval (months)	6

10.2 Troubleshooting HP-210

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
HP-210-E01	abnormal noise	Faulty coolant pump	Replace or reseal coolant pump and retest	P2
HP-210-E02	loss of calibration	Faulty display	Replace or reseal display and retest	P3
HP-210-E03	overheating	Faulty sensor array	Replace or reseal sensor array and retest	P3
HP-210-E04	no response to controls	Faulty gearbox	Replace or reseal gearbox and retest	P4
HP-210-E05	intermittent shutdown	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P2
HP-210-E06	error code on display	Faulty cable harness	Replace or reseal cable harness and retest	P3

11. Product HP-220

Product HP-220 belongs to the Hydraulic Press series. Warranty is 12 months from commissioning.

11.1 Specification HP-220

Parameter	Value
Model	HP-220
Warranty (months)	12
Rated Voltage (V)	230
Service Interval (months)	9

11.2 Troubleshooting HP-220

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
HP-220-E01	loss of calibration	Faulty gearbox	Replace or reseal gearbox and retest	P2
HP-220-E02	overheating	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3
HP-220-E03	no response to controls	Faulty cable harness	Replace or reseal cable harness and retest	P3
HP-220-E04	intermittent shutdown	Faulty encoder	Replace or reseal encoder and retest	P4
HP-220-E05	error code on display	Faulty servo drive	Replace or reseal servo drive and retest	P2
HP-220-E06	abnormal noise	Faulty power supply	Replace or reseal power supply and retest	P3

12. Product HP-310

Product HP-310 belongs to the Hydraulic Press series. Warranty is 18 months from commissioning.

12.1 Specification HP-310

Parameter	Value
Model	HP-310
Warranty (months)	18
Rated Voltage (V)	415
Service Interval (months)	12

12.2 Troubleshooting HP-310

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
HP-310-E01	overheating	Faulty encoder	Replace or reseal encoder and retest	P2
HP-310-E02	no response to controls	Faulty servo drive	Replace or reseal servo drive and retest	P3
HP-310-E03	intermittent shutdown	Faulty power supply	Replace or reseal power supply and retest	P3
HP-310-E04	error code on display	Faulty firmware	Replace or reseal firmware and retest	P4
HP-310-E05	abnormal noise	Faulty coolant pump	Replace or reseal coolant pump and retest	P2
HP-310-E06	loss of calibration	Faulty display	Replace or reseal display and retest	P3

13. Product HP-410

Product HP-410 belongs to the Hydraulic Press series. Warranty is 24 months from commissioning.

13.1 Specification HP-410

Parameter	Value
Model	HP-410
Warranty (months)	24
Rated Voltage (V)	230
Service Interval (months)	15

13.2 Troubleshooting HP-410

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
HP-410-E01	no response to controls	Faulty firmware	Replace or reseal firmware and retest	P2
HP-410-E02	intermittent shutdown	Faulty coolant pump	Replace or reseal coolant pump and retest	P3
HP-410-E03	error code on display	Faulty display	Replace or reseal display and retest	P3
HP-410-E04	abnormal noise	Faulty sensor array	Replace or reseal sensor array and retest	P4
HP-410-E05	loss of calibration	Faulty gearbox	Replace or reseal gearbox and retest	P2
HP-410-E06	overheating	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3

14. Product SM-110

Product SM-110 belongs to the Servo Module series. Warranty is 12 months from commissioning.

14.1 Specification SM-110

Parameter	Value
Model	SM-110
Warranty (months)	12
Rated Voltage (V)	415
Service Interval (months)	6

14.2 Troubleshooting SM-110

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
SM-110-E01	intermittent shutdown	Faulty sensor array	Replace or reseal sensor array and retest	P2
SM-110-E02	error code on display	Faulty gearbox	Replace or reseal gearbox and retest	P3
SM-110-E03	abnormal noise	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3
SM-110-E04	loss of calibration	Faulty cable harness	Replace or reseal cable harness and retest	P4
SM-110-E05	overheating	Faulty encoder	Replace or reseal encoder and retest	P2
SM-110-E06	no response to controls	Faulty servo drive	Replace or reseal servo drive and retest	P3

15. Product SM-120

Product SM-120 belongs to the Servo Module series. Warranty is 18 months from commissioning.

15.1 Specification SM-120

Parameter	Value
Model	SM-120
Warranty (months)	18
Rated Voltage (V)	230
Service Interval (months)	9

15.2 Troubleshooting SM-120

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
SM-120-E01	error code on display	Faulty cable harness	Replace or reseal cable harness and retest	P2
SM-120-E02	abnormal noise	Faulty encoder	Replace or reseal encoder and retest	P3
SM-120-E03	loss of calibration	Faulty servo drive	Replace or reseal servo drive and retest	P3
SM-120-E04	overheating	Faulty power supply	Replace or reseal power supply and retest	P4
SM-120-E05	no response to controls	Faulty firmware	Replace or reseal firmware and retest	P2
SM-120-E06	intermittent shutdown	Faulty coolant pump	Replace or reseal coolant pump and retest	P3

16. Product SM-210

Product SM-210 belongs to the Servo Module series. Warranty is 24 months from commissioning.

16.1 Specification SM-210

Parameter	Value
Model	SM-210
Warranty (months)	24
Rated Voltage (V)	415
Service Interval (months)	12

16.2 Troubleshooting SM-210

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
SM-210-E01	abnormal noise	Faulty power supply	Replace or reseal power supply and retest	P2
SM-210-E02	loss of calibration	Faulty firmware	Replace or reseal firmware and retest	P3
SM-210-E03	overheating	Faulty coolant pump	Replace or reseal coolant pump and retest	P3
SM-210-E04	no response to controls	Faulty display	Replace or reseal display and retest	P4
SM-210-E05	intermittent shutdown	Faulty sensor array	Replace or reseal sensor array and retest	P2
SM-210-E06	error code on display	Faulty gearbox	Replace or reseal gearbox and retest	P3

17. Product SM-220

Product SM-220 belongs to the Servo Module series. Warranty is 12 months from commissioning.

17.1 Specification SM-220

Parameter	Value
Model	SM-220
Warranty (months)	12
Rated Voltage (V)	230
Service Interval (months)	15

17.2 Troubleshooting SM-220

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
SM-220-E01	loss of calibration	Faulty display	Replace or reseal display and retest	P2
SM-220-E02	overheating	Faulty sensor array	Replace or reseal sensor array and retest	P3
SM-220-E03	no response to controls	Faulty gearbox	Replace or reseal gearbox and retest	P3
SM-220-E04	intermittent shutdown	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P4
SM-220-E05	error code on display	Faulty cable harness	Replace or reseal cable harness and retest	P2
SM-220-E06	abnormal noise	Faulty encoder	Replace or reseal encoder and retest	P3

18. Product SM-310

Product SM-310 belongs to the Servo Module series. Warranty is 18 months from commissioning.

18.1 Specification SM-310

Parameter	Value
Model	SM-310
Warranty (months)	18
Rated Voltage (V)	415
Service Interval (months)	6

18.2 Troubleshooting SM-310

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
SM-310-E01	overheating	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P2
SM-310-E02	no response to controls	Faulty cable harness	Replace or reseal cable harness and retest	P3
SM-310-E03	intermittent shutdown	Faulty encoder	Replace or reseal encoder and retest	P3
SM-310-E04	error code on display	Faulty servo drive	Replace or reseal servo drive and retest	P4
SM-310-E05	abnormal noise	Faulty power supply	Replace or reseal power supply and retest	P2
SM-310-E06	loss of calibration	Faulty firmware	Replace or reseal firmware and retest	P3

19. Product SM-410

Product SM-410 belongs to the Servo Module series. Warranty is 24 months from commissioning.

19.1 Specification SM-410

Parameter	Value
Model	SM-410
Warranty (months)	24
Rated Voltage (V)	230
Service Interval (months)	9

19.2 Troubleshooting SM-410

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
SM-410-E01	no response to controls	Faulty servo drive	Replace or reseal servo drive and retest	P2
SM-410-E02	intermittent shutdown	Faulty power supply	Replace or reseal power supply and retest	P3
SM-410-E03	error code on display	Faulty firmware	Replace or reseal firmware and retest	P3
SM-410-E04	abnormal noise	Faulty coolant pump	Replace or reseal coolant pump and retest	P4
SM-410-E05	loss of calibration	Faulty display	Replace or reseal display and retest	P2
SM-410-E06	overheating	Faulty sensor array	Replace or reseal sensor array and retest	P3

20. Product TL-110

Product TL-110 belongs to the Tooling Line series. Warranty is 12 months from commissioning.

20.1 Specification TL-110

Parameter	Value
Model	TL-110
Warranty (months)	12
Rated Voltage (V)	415
Service Interval (months)	12

20.2 Troubleshooting TL-110

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
TL-110-E01	intermittent shutdown	Faulty coolant pump	Replace or reseal coolant pump and retest	P2
TL-110-E02	error code on display	Faulty display	Replace or reseal display and retest	P3
TL-110-E03	abnormal noise	Faulty sensor array	Replace or reseal sensor array and retest	P3
TL-110-E04	loss of calibration	Faulty gearbox	Replace or reseal gearbox and retest	P4
TL-110-E05	overheating	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P2
TL-110-E06	no response to controls	Faulty cable harness	Replace or reseal cable harness and retest	P3

21. Product TL-120

Product TL-120 belongs to the Tooling Line series. Warranty is 18 months from commissioning.

21.1 Specification TL-120

Parameter	Value
Model	TL-120
Warranty (months)	18
Rated Voltage (V)	230
Service Interval (months)	15

21.2 Troubleshooting TL-120

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
TL-120-E01	error code on display	Faulty gearbox	Replace or reseal gearbox and retest	P2
TL-120-E02	abnormal noise	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3
TL-120-E03	loss of calibration	Faulty cable harness	Replace or reseal cable harness and retest	P3
TL-120-E04	overheating	Faulty encoder	Replace or reseal encoder and retest	P4
TL-120-E05	no response to controls	Faulty servo drive	Replace or reseal servo drive and retest	P2
TL-120-E06	intermittent shutdown	Faulty power supply	Replace or reseal power supply and retest	P3

22. Product TL-210

Product TL-210 belongs to the Tooling Line series. Warranty is 24 months from commissioning.

22.1 Specification TL-210

Parameter	Value
Model	TL-210
Warranty (months)	24
Rated Voltage (V)	415
Service Interval (months)	6

22.2 Troubleshooting TL-210

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
TL-210-E01	abnormal noise	Faulty encoder	Replace or reseal encoder and retest	P2
TL-210-E02	loss of calibration	Faulty servo drive	Replace or reseal servo drive and retest	P3
TL-210-E03	overheating	Faulty power supply	Replace or reseal power supply and retest	P3
TL-210-E04	no response to controls	Faulty firmware	Replace or reseal firmware and retest	P4
TL-210-E05	intermittent shutdown	Faulty coolant pump	Replace or reseal coolant pump and retest	P2
TL-210-E06	error code on display	Faulty display	Replace or reseal display and retest	P3

23. Product TL-220

Product TL-220 belongs to the Tooling Line series. Warranty is 12 months from commissioning.

23.1 Specification TL-220

Parameter	Value
Model	TL-220
Warranty (months)	12
Rated Voltage (V)	230
Service Interval (months)	9

23.2 Troubleshooting TL-220

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
TL-220-E01	loss of calibration	Faulty firmware	Replace or reseal firmware and retest	P2
TL-220-E02	overheating	Faulty coolant pump	Replace or reseal coolant pump and retest	P3
TL-220-E03	no response to controls	Faulty display	Replace or reseal display and retest	P3
TL-220-E04	intermittent shutdown	Faulty sensor array	Replace or reseal sensor array and retest	P4
TL-220-E05	error code on display	Faulty gearbox	Replace or reseal gearbox and retest	P2
TL-220-E06	abnormal noise	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3

24. Product TL-310

Product TL-310 belongs to the Tooling Line series. Warranty is 18 months from commissioning.

24.1 Specification TL-310

Parameter	Value
Model	TL-310
Warranty (months)	18
Rated Voltage (V)	415
Service Interval (months)	12

24.2 Troubleshooting TL-310

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
TL-310-E01	overheating	Faulty sensor array	Replace or reseal sensor array and retest	P2
TL-310-E02	no response to controls	Faulty gearbox	Replace or reseal gearbox and retest	P3
TL-310-E03	intermittent shutdown	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3
TL-310-E04	error code on display	Faulty cable harness	Replace or reseal cable harness and retest	P4
TL-310-E05	abnormal noise	Faulty encoder	Replace or reseal encoder and retest	P2
TL-310-E06	loss of calibration	Faulty servo drive	Replace or reseal servo drive and retest	P3

25. Product TL-410

Product TL-410 belongs to the Tooling Line series. Warranty is 24 months from commissioning.

25.1 Specification TL-410

Parameter	Value
Model	TL-410
Warranty (months)	24
Rated Voltage (V)	230
Service Interval (months)	15

25.2 Troubleshooting TL-410

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
TL-410-E01	no response to controls	Faulty cable harness	Replace or reseal cable harness and retest	P2
TL-410-E02	intermittent shutdown	Faulty encoder	Replace or reseal encoder and retest	P3
TL-410-E03	error code on display	Faulty servo drive	Replace or reseal servo drive and retest	P3
TL-410-E04	abnormal noise	Faulty power supply	Replace or reseal power supply and retest	P4
TL-410-E05	loss of calibration	Faulty firmware	Replace or reseal firmware and retest	P2
TL-410-E06	overheating	Faulty coolant pump	Replace or reseal coolant pump and retest	P3

26. Product VX-110

Product VX-110 belongs to the Vision Unit series. Warranty is 12 months from commissioning.

26.1 Specification VX-110

Parameter	Value
Model	VX-110
Warranty (months)	12
Rated Voltage (V)	415
Service Interval (months)	6

26.2 Troubleshooting VX-110

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
VX-110-E01	intermittent shutdown	Faulty power supply	Replace or reseal power supply and retest	P2
VX-110-E02	error code on display	Faulty firmware	Replace or reseal firmware and retest	P3
VX-110-E03	abnormal noise	Faulty coolant pump	Replace or reseal coolant pump and retest	P3
VX-110-E04	loss of calibration	Faulty display	Replace or reseal display and retest	P4
VX-110-E05	overheating	Faulty sensor array	Replace or reseal sensor array and retest	P2
VX-110-E06	no response to controls	Faulty gearbox	Replace or reseal gearbox and retest	P3

27. Product VX-120

Product VX-120 belongs to the Vision Unit series. Warranty is 18 months from commissioning.

27.1 Specification VX-120

Parameter	Value
Model	VX-120
Warranty (months)	18
Rated Voltage (V)	230
Service Interval (months)	9

27.2 Troubleshooting VX-120

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
VX-120-E01	error code on display	Faulty display	Replace or reseal display and retest	P2
VX-120-E02	abnormal noise	Faulty sensor array	Replace or reseal sensor array and retest	P3
VX-120-E03	loss of calibration	Faulty gearbox	Replace or reseal gearbox and retest	P3
VX-120-E04	overheating	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P4
VX-120-E05	no response to controls	Faulty cable harness	Replace or reseal cable harness and retest	P2
VX-120-E06	intermittent shutdown	Faulty encoder	Replace or reseal encoder and retest	P3

28. Product VX-210

Product VX-210 belongs to the Vision Unit series. Warranty is 24 months from commissioning.

28.1 Specification VX-210

Parameter	Value
Model	VX-210
Warranty (months)	24
Rated Voltage (V)	415
Service Interval (months)	12

28.2 Troubleshooting VX-210

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
VX-210-E01	abnormal noise	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P2
VX-210-E02	loss of calibration	Faulty cable harness	Replace or reseal cable harness and retest	P3
VX-210-E03	overheating	Faulty encoder	Replace or reseal encoder and retest	P3
VX-210-E04	no response to controls	Faulty servo drive	Replace or reseal servo drive and retest	P4
VX-210-E05	intermittent shutdown	Faulty power supply	Replace or reseal power supply and retest	P2
VX-210-E06	error code on display	Faulty firmware	Replace or reseal firmware and retest	P3

29. Product VX-220

Product VX-220 belongs to the Vision Unit series. Warranty is 12 months from commissioning.

29.1 Specification VX-220

Parameter	Value
Model	VX-220
Warranty (months)	12
Rated Voltage (V)	230
Service Interval (months)	15

29.2 Troubleshooting VX-220

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
VX-220-E01	loss of calibration	Faulty servo drive	Replace or reseal servo drive and retest	P2
VX-220-E02	overheating	Faulty power supply	Replace or reseal power supply and retest	P3
VX-220-E03	no response to controls	Faulty firmware	Replace or reseal firmware and retest	P3
VX-220-E04	intermittent shutdown	Faulty coolant pump	Replace or reseal coolant pump and retest	P4
VX-220-E05	error code on display	Faulty display	Replace or reseal display and retest	P2
VX-220-E06	abnormal noise	Faulty sensor array	Replace or reseal sensor array and retest	P3

30. Product VX-310

Product VX-310 belongs to the Vision Unit series. Warranty is 18 months from commissioning.

30.1 Specification VX-310

Parameter	Value
Model	VX-310
Warranty (months)	18
Rated Voltage (V)	415
Service Interval (months)	6

30.2 Troubleshooting VX-310

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
VX-310-E01	overheating	Faulty coolant pump	Replace or reseal coolant pump and retest	P2
VX-310-E02	no response to controls	Faulty display	Replace or reseal display and retest	P3
VX-310-E03	intermittent shutdown	Faulty sensor array	Replace or reseal sensor array and retest	P3
VX-310-E04	error code on display	Faulty gearbox	Replace or reseal gearbox and retest	P4
VX-310-E05	abnormal noise	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P2
VX-310-E06	loss of calibration	Faulty cable harness	Replace or reseal cable harness and retest	P3

31. Product VX-410

Product VX-410 belongs to the Vision Unit series. Warranty is 24 months from commissioning.

31.1 Specification VX-410

Parameter	Value
Model	VX-410
Warranty (months)	24
Rated Voltage (V)	230
Service Interval (months)	9

31.2 Troubleshooting VX-410

Fault Code	Symptom	Probable Cause	Recommended Action	Priority
VX-410-E01	no response to controls	Faulty gearbox	Replace or reseal gearbox and retest	P2
VX-410-E02	intermittent shutdown	Faulty hydraulic valve	Replace or reseal hydraulic valve and retest	P3
VX-410-E03	error code on display	Faulty cable harness	Replace or reseal cable harness and retest	P3
VX-410-E04	abnormal noise	Faulty encoder	Replace or reseal encoder and retest	P4
VX-410-E05	loss of calibration	Faulty servo drive	Replace or reseal servo drive and retest	P2
VX-410-E06	overheating	Faulty power supply	Replace or reseal power supply and retest	P3